

4.1.

$$\text{Chci aby } \sum_i (z_i - (ax_i + by_i + c))^2$$

$$\text{byla min. tedy } \sum_i (z_i - (x_i y_i 1) \begin{pmatrix} a \\ b \\ c \end{pmatrix})^2 =$$

$$= \sum_i \left((x_i y_i 1) \underset{\hat{x}}{\begin{pmatrix} a \\ b \\ c \end{pmatrix}} - z_i \right)^2 = \hat{x} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

$$= \|A\hat{x} - \vec{z}\|, \text{ kde } A \text{ je}$$

$$\begin{pmatrix} x_1 & y_1 & 1 \\ \vdots & \vdots & \vdots \\ x_6 & y_6 & 1 \end{pmatrix}$$

$$\begin{pmatrix} x_i & y_i & 1 & z_i \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & -1 & 1 & 0 \\ -1 & 1 & 1 & 1 \end{pmatrix}$$

Tedy použijí
metodu nejmenší
čtverců

$$G[\hat{x}]_B = A^* \vec{z}$$

$$G = \begin{pmatrix} 4 & -1 & 2 \\ -1 & 4 & 2 \\ 2 & 2 & 6 \end{pmatrix}$$

$$A^* \vec{z} = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$$

$$\left(\begin{array}{ccc|c} -1 & 4 & 2 & 4 \\ 2 & 2 & 6 & 6 \\ 4 & -1 & 2 & 2 \end{array} \right) \rightarrow \begin{pmatrix} 2/5 \\ 4/5 \\ 3/5 \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

$$\text{rovina tedy bude } \underline{\underline{z = \frac{2}{5}x + \frac{4}{5}y + \frac{3}{5}}}$$